C30 Amino Acids, Proteins and DNA – Knowledge Organiser

AQA A-level Chemistry • Set 1: Amino Acids and Proteins **Paper 2**

#	Question	Answer
1	Functional groups in amino acids	$-NH_2$ and –COOH
2	Zwitterion	Ion with both + and − charges
3	Amino acid in acidic solution	$-NH_3^+$ form
4	Amino acid in alkaline solution	$-COO^-$ form
5	Why amino acids have high melting points	Ionic attraction between zwitterions
6	Peptide link	–CONH–
7	Primary structure	Sequence of amino acids
8	Secondary structure	α-helix or β-pleated sheet (hydrogen bonds)
9	Tertiary structure	3D folding (H bonds, ionic bonds, disulfide bridges)
10	Conditions to hydrolyse a protein	6 mol dm^{-3} HCl, reflux

C30 Amino Acids, Proteins and DNA – Knowledge Organiser

AQA A-level Chemistry • Set 2: Enzymes and Chromatography **Paper 2**

#	Question	Answer
1	Amino acid that forms disulfide bridges	Cysteine
2	Disulfide bridge	S–S bond
3	Achiral amino acid	Glycine
4	Enzyme active site	Stereospecific
5	Enzyme inhibitor	Binds to the active site, blocking the substrate
6	Use of computer modelling in drug design	Designs molecules to fit an active site
7	Locating agent for amino acids	Ninhydrin
8	R_f equation	Distance moved by spot ÷ distance moved by solvent
9	How amino acids are identified by TLC	Compare R_f values
10	Bond broken in protein hydrolysis	Peptide (amide) bond

C30 Amino Acids, Proteins and DNA – Knowledge Organiser

AQA A-level Chemistry • Set 3: DNA **Paper 2**

#	Question	Answer
1	Monomer of DNA	Nucleotide
2	Three parts of a nucleotide	Phosphate, 2-deoxyribose, base
3	Four bases in DNA	Adenine, cytosine, guanine, thymine
4	Complementary base pairs	A–T and C–G
5	Hydrogen bonds between A and T	2
6	Hydrogen bonds between C and G	3
7	Shape of DNA	Double helix
8	DNA backbone	Sugar–phosphate
9	How cisplatin works	Binds to N on guanine; stops DNA replication
10	Side effect of cisplatin	Also damages healthy cells